

Concordia

Federated learning on harmonised biobank data

SUBPROJECT 1

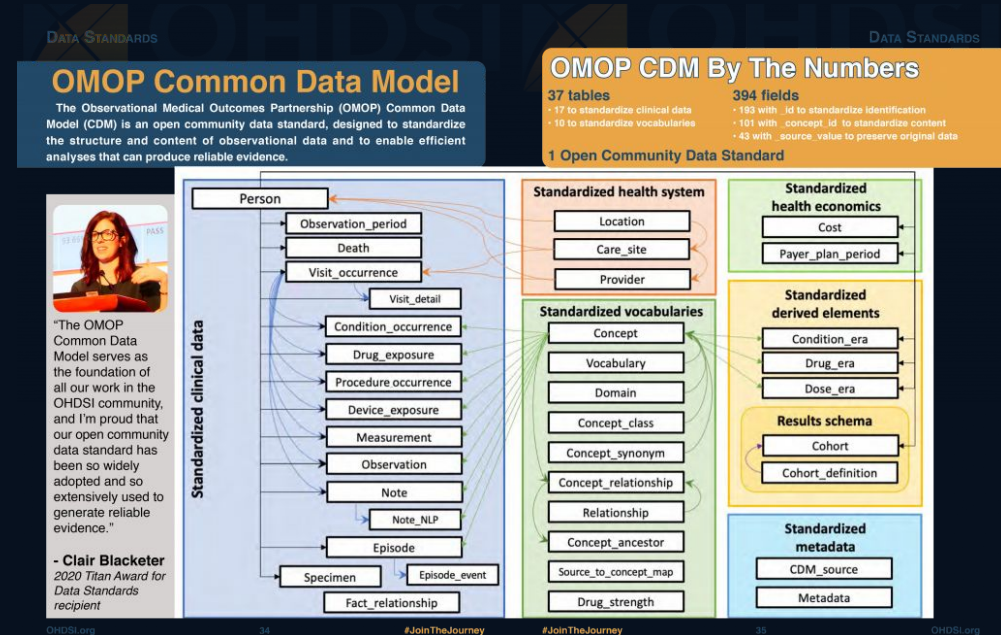
Agentic conversion to OMOP

SUBPROJECT 2

Scalable federated learning on OMOP data
using NVFlare

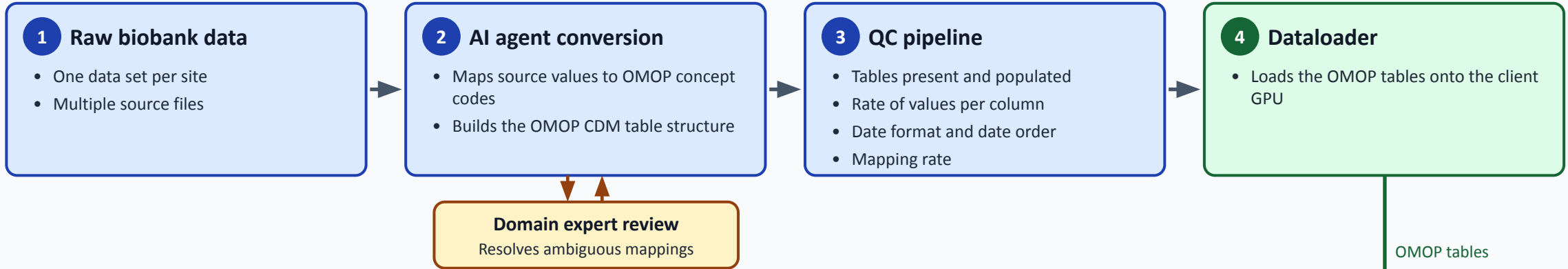
Motivation

- Biobank data often differ substantially between sites
- Manual mapping is labour intensive and not scalable
- Goal: Automated conversion framework to standardize heterogeneous biobank datasets into OMOP.

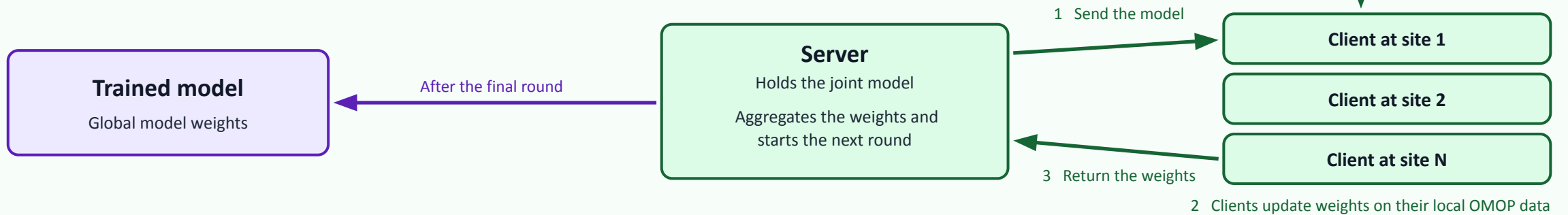


Concordia: from raw biobank data to a federated model

Repeated at each site



Federated training loop (NVFlare)



Hackathon prototype: two end-to-end runs on simulated sites and one proof-of-concept in a real world setting

Synthea synthetic data

5 simulated sites

Synthetic UK Biobank data

8 simulated sites

HUNT cloud

Proof-of-concept
connection via AWS

Site simulation

- New omopflare framework provides building blocks to load data efficiently into GPU memory.
- NVFlare simulates both the server and clients to enable federated learning.

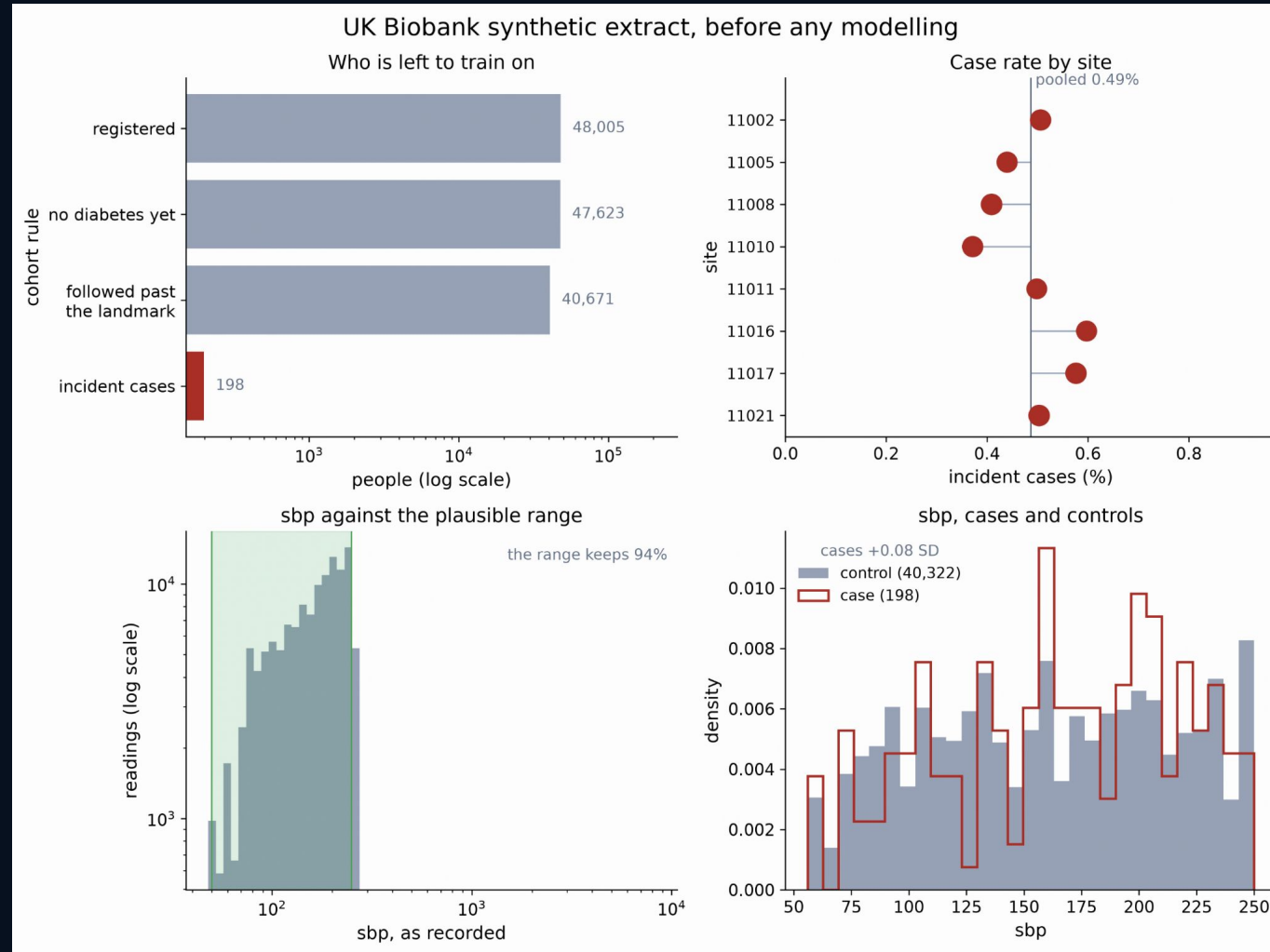
A skill to map biobank data onto OMOP



Two OMOP mapping approaches

- both are LLM-assisted mapping using DuckDB
- 1) Fully autonomous, no Athena lookup during execution.
 - 2) Coded rule-based mapping using Athena OMOP vocabulary, with user interaction

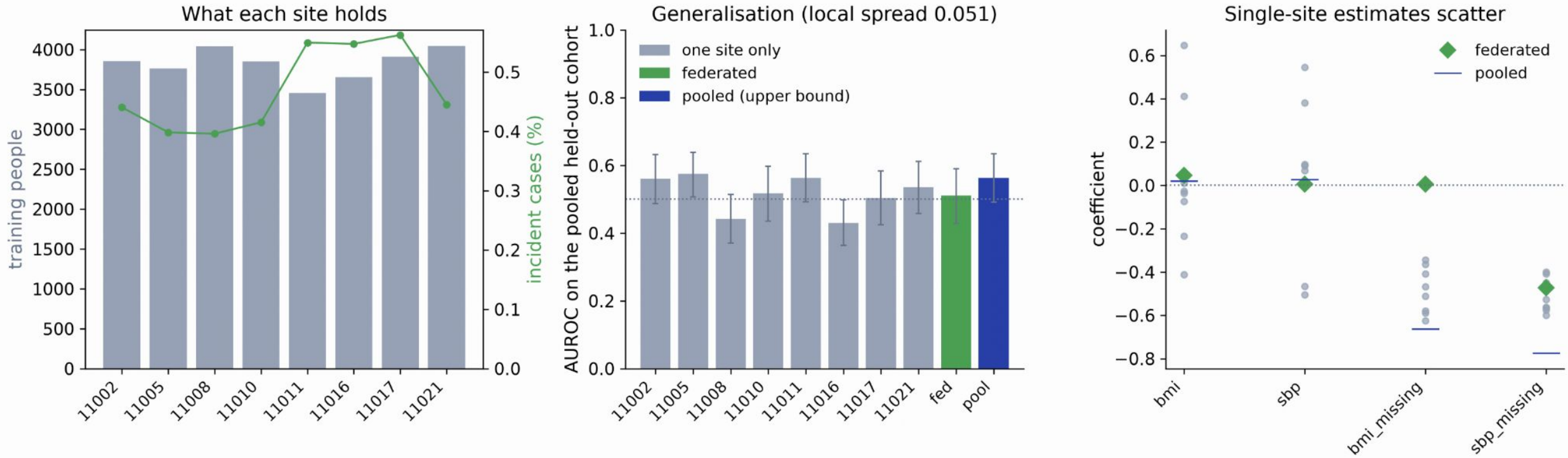
Overview of UKB cohort - diabetes use case



Federated learning on simulated UKB data

Results aren't meaningful by design - but it's a proof of concept

Incident type 2 diabetes across UK Biobank assessment centres



Future steps

- Validating the OMOP skill on additional real world biobank datasets
 - Hunt
 - All of Us
 - MIMIK
- Investigate how to best agree on schema between clients
- Scaling the dataloader beyond 1 billion patient visits
- Publish omopflare framework on pypi
- Adding a OMOP use case to the official NVFlare documentation
- prototyping a sippable on site Version of our ETL

